

ANKIT KUMAR

ankitkumar.itbhu@gmail.com • +1 (857) 869-5430
ankitku.github.io • linkedin.com/in/ankitkumar1988 • github.com/ankitku
Northeastern University, Boston, MA

PROFESSIONAL SUMMARY

Applied Scientist and PhD candidate with demonstrated expertise in formal methods, security analysis, and software verification. Discovered and reported CVE-2022-47547 affecting Ethereum 2.0 networks, resulting in protocol-level patches. Author of 7 peer-reviewed publications (32 citations, h-index: 4) with 2 Best Student Paper Awards. Holds 3 pending patents from Amazon internships. Recognized internationally through invited presentations at IPFS Camp Lisbon and multiple ACL2 Workshops, with peer review service for premier venues (FMCAD, ICFEM, NFM, OOPSLA).

TECHNICAL SKILLS

- **Programming Languages:** C, C++, Java, Rust, Lisp, ACL2, ACL2 Sedan, Coq, Haskell, Ruby
- **Development Tools:** Git, GitHub, Emacs, LaTeX, Shell Scripting, Perforce
- **Frameworks & Technologies:** Spring, .NET, Grails, Ruby on Rails, Android
- **AI & Machine Learning:** Claude, Cline, Amazon Q, Agentic AI
- **Verification Tools:** Jaspergold, ACL2, ACL2 Sedan
- **Specializations:** Formal Methods, Theorem Proving, AI Agents, Software Verification, Security Analysis, Dataflow Analysis, Taint Analysis

PROFESSIONAL EXPERIENCE

Applied Scientist Intern • Amazon Web Services • Santa Clara, CA • May 19, 2025 - Aug 22, 2025

- Pioneered **Agentic AI methodology** to automatically generate correct Rust code specifications from Java implementations
- Leveraged advanced **dataflow analysis** to eliminate hallucinations and improve code correctness
- Secured **two patent submissions (pending)** for innovative code verification and analysis techniques; earned return offer recognition in a full-time role
- Technologies: Java, Rust, Claude, Cline, Amazon Q, Agentic AI

Member of Technical Staff, Intern • Rivos Inc. • Portland, OR • Jun 7, 2023 - Sep 1, 2023

- Developed formal verification framework for **RISC-V hypervisor page walk algorithm security properties**
- Accelerated proof verification by 10x using strategic lemma transfer from high-level models
- Technologies: ACL2, ACL2 Sedan, Jaspergold

Applied Scientist Intern • Amazon • Minneapolis, MN • May 31, 2022 - Sep 2, 2022

- Engineered **taint-analysis framework** for enterprise-scale tracking of sensitive data across AWS infrastructure
- Secured **one patent submission (pending):** "Data Flow Analysis of Cloud-Based Software Applications"; earned return offer recognition
- Technologies: Java, Grails, GitHub, Shell Scripting

Software Development Engineer II • Olacabs • Bengaluru, India • 2012-2014

- Architected inaugural Olacabs Android consumer application, capturing **80% of total cab bookings**
- Reduced driver app crashes by **80%** through systematic debugging and optimization
- Implemented robust cashless payment and digital wallet systems
- Technologies: Java, Android, Git, Ruby on Rails

Associate • Morgan Stanley Advantage Services • Mumbai, India • 2010-2012

- Transformed legacy derivatives trading platform for multi-regional deployment
- Expanded platform reach to Hong Kong and Singapore, **doubling transaction capacity**
- Technologies: .NET, Java, Spring, Perforce

Upcoming Role: Applied Scientist • Amazon Web Services • Santa Clara, CA • Starting Jan 2026

EDUCATION

PhD in Computer Science • Northeastern University • Boston, MA • 2018-Expected 2026

- Thesis: Refinement-based reasoning of P2P protocols is feasible and useful
- **Research Impact:** 7 peer-reviewed publications, 32 citations, h-index: 4, i10-index: 1

PhD Research Contributions

- **1. P2P Protocol Verification**
 - Applied formal methods and refinement theory to **Gossipsub/Floodsub protocols**
 - Discovered and reported [**MITRE CVE-2022-47547**], a critical scoring function vulnerability affecting Ethereum and Filecoin networks, leading to real-world fixes
 - Published multiple peer-reviewed papers and presented work at IPFS Camp Lisbon 2022 and ACL2 Workshops (2023, 2025)
 - Related publications: [1], [2], [3]
- **2. Education: Automated Student Feedback Systems**
 - Developed **theorem-prover-backed automated feedback systems** for grading student submissions
 - Integrated with Gradescope to provide formal verification of student solutions and proofs
 - Enabled scalable and precise assessment of student work in formal methods courses
 - Related publications: [4], [5], [6]
- **3. Verification tools and algorithms: String Solvers**
 - Designed and implemented sound and theoretically complete **string equation solvers** for verification and formal reasoning tasks
 - Applied these solvers to both academic research and practical software verification problems
 - Related publications: [7]

M.Tech in Computer Science • IIT Kanpur • Kanpur, India • 2015-2017

B.Tech in Electrical Engineering • IIT (BHU) Varanasi • Varanasi, India • 2006-2010

SECURITY RESEARCH & IMPACT

- **CVE-2022-47547 Discovery:** Identified critical vulnerability in GossipSub 1.1 protocol affecting Ethereum 2.0 networks
 - CVSS Score: 5.3/10 (Medium Severity) - Network-based attack, Low Complexity
 - Published: December 19, 2022 — NVD Last Modified: April 17, 2025

- **Impact:** Affects integrity of peer-to-peer message propagation in Ethereum consensus layer
- Coordinated disclosure with MITRE; resulted in protocol-level patches deployed globally
- **Formal Verification Research:** Applied mathematical proof techniques to analyze Floodsub and Gossipsub protocols, revealing previously unknown attack vectors
- **Publications:** 7 peer-reviewed papers (32 citations) demonstrating formal verification methodologies for distributed systems
- **Patent Portfolio:** 3 patent applications pending with US Patent Office covering verification and security analysis techniques

PROFESSIONAL SERVICE & COMMUNITY CONTRIBUTIONS _____

- **Peer Review Service:** Reviewed papers for FMCAD 2025, ICFEM 2023, and NFM 2024 (premier formal methods conferences)
- **Artifacts Evaluation:** Served on OOPSLA 2024 and ICFP 2023 Artifacts Evaluation Committees (top-tier programming languages conferences)
- **Conference Leadership:** Invited to chair SIGCSE TS Conference, Canada, 2023 (leading computer science education conference)
- **Volunteer Service:** Student volunteer at FLoC 2022, Haifa, Israel (Federated Logic Conferences)

PUBLICATIONS _____

1. **A Formalization of the Correctness of the Floodsub Protocol**
ACL2 Workshop 2025, Austin, TX • Best Student Paper Award • [\[PDF\]](#)
2. **Formal Model-Driven Analysis of Resilience of GossipSub to Attacks from Misbehaving Peers**
IEEE Symposium on Security and Privacy 2024, San Francisco, CA • [\[PDF\]](#)
3. **Verification of GossipSub in ACL2s**
ACL2 Workshop 2023, Austin, TX • Best Student Paper Award • [\[PDF\]](#)
4. **Automated Grading of Automata with ACL2s**
Electronic Proceedings in Theoretical Computer Science 2023 • [\[PDF\]](#)
5. **Proving Calculational Proofs Correct**
ACL2 Workshop 2023, Austin, TX • [\[PDF\]](#)
6. **Calculational Proofs in ACL2s**
arXiv preprint 2023 • [\[PDF\]](#)
7. **Mathematical Programming Modulo Strings**
Formal Methods in Computer Aided Design 2021, Virtual Conference • [\[PDF\]](#)

AWARDS & HONORS _____

- Best Student Paper Award, ACL2 Workshop 2025
- Dissertation Completion Fellowship, Summer 2024
- Invited Talk, Syracuse University 2024
- Best Student Paper Award, ACL2 Workshop 2023
- Invited Talk, IPFS Camp Lisbon 2022

- Student Travel Fellowship, PLMW at POPL 2019
- First Prize, Droidcon India Hackathon 2013